



ISO/TC 12 Quantities and units

Report of the Secretariat to the meeting of ISO/TC 12 to be held virtual via Zoom, May 19th, 2022.

Year of creation	1947
Secretariat	SIS, Sweden Since 1982 the secretariat is held by SIS.
Chairperson	Dr Torgny Carlsson (until end 2024)
Committee Manager	Mr Sven Radhe, SIS
Secretariat support	Ms Anette Lindor Norén, SIS
ISO Technical Programme Manager	Ms Mercè Ferrés Hernandez
ISO Editorial Manager	Mr Vincenzo Bazzucchi
Scope of ISO/TC 12	Standardization of units and symbols for quantities and units (and mathematical symbols) used within the different fields of science and technology, giving, where necessary, definitions of these quantities and units. Standard conversion factors between the various units.

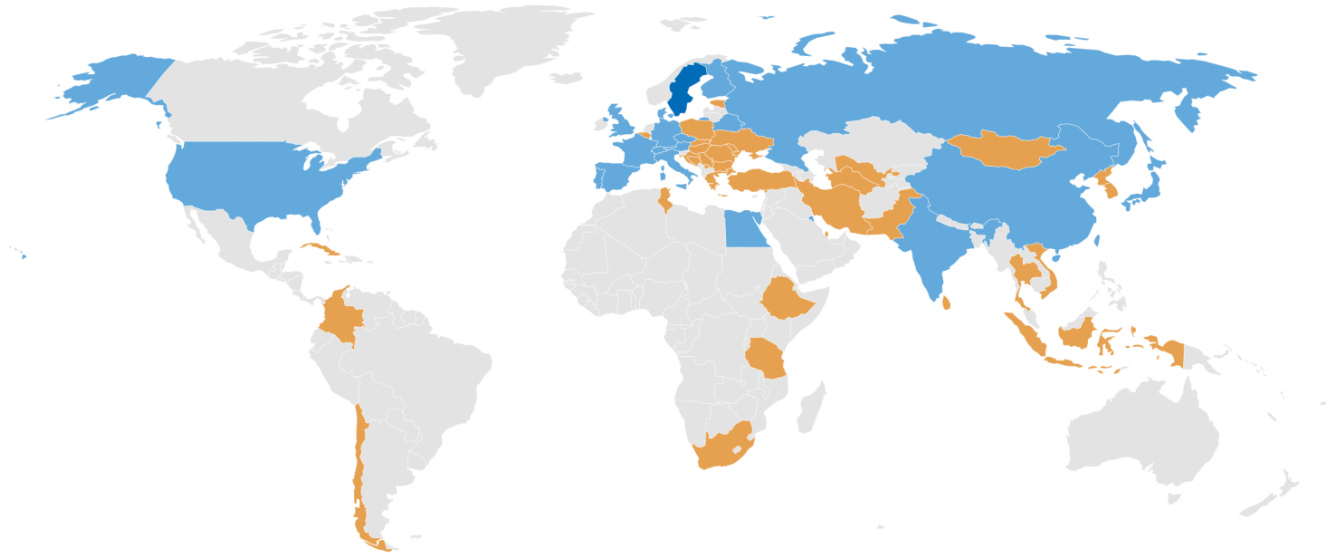
ISO/TC 12 contributes to the following Sustainable Development Goals:

Goal 4: Quality education: Ensure inclusive and equitable quality education and promote lifelong learning opportunities for all

Goal 9: Industry, innovation and infrastructure: Build resilient infrastructure, promote inclusive and sustainable industrialization and foster innovation



Members 2022-05-10



This map is designed to visually demonstrate the geographic distribution of our Members. The boundaries shown do not imply an official endorsement or acceptance by ISO.

● Participating Countries (20)

- Austria (ASI)
- Belarus (BELST)
- China (SAC)
- Czech Republic (UNMZ)
- Denmark (DS)
- Egypt (EOS)
- Finland (SFS)
- France (AFNOR)
- Germany (DIN)
- India (BIS)
- Italy (UNI)
- Japan (JISC)
- Kuwait (KOWSMD)
- Portugal (IPQ)
- Russian Federation (GOST R)
- Spain (UNE)
- Sweden (SIS)
- Switzerland (SNV)
- United Kingdom (BSI)
- United States (ANSI)

NOTE *P-members* (participating members) participate actively in the work, with an obligation to vote on all questions formally submitted for voting within the TC or SC, on new work item proposals, enquiry drafts and final draft International Standards, and to contribute to meetings.
O-members (observers) follow the work as an observer, and therefore receive committee documents and have the right to submit comments and to attend meetings.

● Observing Countries (35)

- Belgium (NBN)
- Bosnia and Herzegovina (BAS)
- Bulgaria (BDS)
- Chile (INN)
- Colombia (ICONTEC)
- Croatia (HZN)
- Cuba (NC)
- Estonia (EVS)
- Ethiopia (ESA)
- Greece (NQIS ELOT)
- Hong Kong (ITCHK SAR)
- Hungary (MSZT)
- Indonesia (BSN)
- Iran, Islamic Republic of (ISIRI)
- Korea, Democratic People's Republic of (CSK)
- Korea, Republic of (KATS)
- Mauritius (MSB)
- Moldova, Republic of (ISM)
- Mongolia (MASM)
- Pakistan (PSQCA)
- Poland (PKN)
- Qatar (QS)
- Romania (ASRO)
- Serbia (ISS)
- Slovakia (SOSMT)
- South Africa (SABS)
- Sri Lanka (SLSI)
- Tanzania, United Republic of (TBS)
- Thailand (TISI)
- Tunisia (INNORPI)
- Turkey (TSE)
- Turkmenistan (MSST)
- Ukraine (DSTU)
- Uzbekistan (UZSTANDARD)
- Viet Nam (STAMEQ)



Liaison Committees from ISO/TC 12

ISO/TC 12 can access the documents of the committees below:

Reference	Title	Liaison representative
CIE	International Commission on Illumination	Ms. Lindor Norén, Anette Mr. Radhe, Sven
IEC/TC 1	Terminology	Mr. Radhe, Sven
IEC/TC 25	Quantities and units	Dr. Ehrlich, Charles D
ISO/IEC JTC 1/SC 32	Data management and interchange	Ms. Lindor Norén, Anette Mr. Radhe, Sven
ISO/TC 8/SC 6	Navigation and ship operations	Ms. Lindor Norén, Anette Mr. Radhe, Sven
ISO/TC 37	Language and terminology	Ms. Lindor Norén, Anette Mr. Radhe, Sven
ISO/TC 43	Acoustics	Ms. Lindor Norén, Anette Mr. Radhe, Sven
ISO/TC 43/SC 1	Noise	Ms. Lindor Norén, Anette Mr. Radhe, Sven
ISO/TC 47	Chemistry	Ms. Lindor Norén, Anette Mr. Radhe, Sven
ISO/TC 69	Applications of statistical methods	Ms. Lindor Norén, Anette Mr. Radhe, Sven
ISO/TC 85	Nuclear energy, nuclear technologies, and radiological protection	Ms. Lindor Norén, Anette Mr. Radhe, Sven
ISO/TC 115	Pumps	Ms. Lindor Norén, Anette Mr. Radhe, Sven
ISO/TC 215	Health informatics	Dr. Cullati, Stéphane Prof. Pendrill, Leslie



Liaison Committee to ISO/TC 12

The committees and organisations below can access the documents of ISO/TC 12:

Reference	Title	Liaison representative
CIE	International Commission on Illumination	Dr Islam, Shahidul Ms Nield, Kathryn Dr Zwick, Peter
IEC/TC 25	Quantities and units	Prof. Mari, Luca
ISO/IEC JTC 1/SC 32	Data management and interchange	Ms. Miller, Michaela Ms Rajchel, Lisa Mr Wennblom, Phil
ISO/TC 37	Language and terminology	Mr Cheng, Yonghong Dr Galinski, Christian Mr Wang, Haitao Mr Zhou, Changqing
ISO/TC 43	Acoustics	Mr Kornprobst, Hans-Joachim Ms Kursch, Stephanie Ms Liebing, Daniela
ISO/TC 43/SC 1	Noise	Mr Kornprobst, Hans-Joachim Ms Kursch, Stephanie Ms Liebing, Daniela
ISO/TC 43/SC 3	Underwater acoustics	Mr Ainslie, Michael
ISO/TC 47	Chemistry	Ms Nonaka, Reiko
ISO/TC 69	Applications of statistical methods	Ms Admussen, Jennifer
ISO/TC 85	Nuclear energy, nuclear technologies, and radiological protection	Mr Balcaen, Eric
ISO/TC 215	Health informatics	Dr Geßner, Christof Ms Stanford, Sharon Ms Warner, Diana



Organizations in liaison (Category A and B)
ISO/TC 12 has liaisons with the following organisations:

Acronym	Title	Category	Liaison representative
BIPM	International Bureau of Weights and Measures	A	Dr Milton, Martin Mr Sitton, Robert
EC - European Commission	European Commission	B	
EURAMET	European Association of National Metrology Institutes	A	Mr Jarvis, Duncan
IAEA	International Atomic Energy Agency	A	
ICAO	International Civil Aviation Organization	A	
ICRU	International Commission on Radiation Units and Measurements	A	Dr Schauer, David A.
IMO	International Maritime Organization	B	Ms Allnutt, Sandra Rita
Infoterm	International Information Centre for Terminology (Infoterm)	A	Dr Galinski, Christian
ITU	International Telecommunication Union	A	Mr Caras, John George Ms Xiaoya Yang, xiaoya (ITU-T SG 17)
IUCr	International Union of Crystallography	A	Prof Brock, C.P. Mr Dacombe, Michael
IUPAC	International Union of Pure and Applied Chemistry	A	Prof Mills, Ian
IUPAP	International Union of Pure and Applied Physics	A	Dr Karshenboim, Savely Prof Pendrill, Leslie
OASIS	Organization for the Advancement of Structured Information Standards (OASIS)	A	Mr Olken, Frank
OIML	International Organization of Legal Metrology	A	Mr Goblirsch, Richard



UNECE	United Nations Economic Commission for Europe	A	Mr Henninot, Jean-Pierre
WCO	World Customs Organization	A	
WHO	World Health Organization	A	Dr Ustun, Bedirhan
WMO	World Meteorological Organization	A	Mr Shi, Peiliang

Category A: Organizations that make an effective contribution to the work of the technical committee or subcommittee for questions dealt with by this technical committee or subcommittee. Such organizations are given access to all relevant documentation and are invited to meetings. They may nominate Experts to participate in a WG (see 1.12.1).

Category B: Organizations that have indicated a wish to be kept informed of the work of the technical committee or subcommittee. Such organizations are given access to reports on the work of a technical committee or subcommittee. Category B is reserved for inter-governmental organizations.

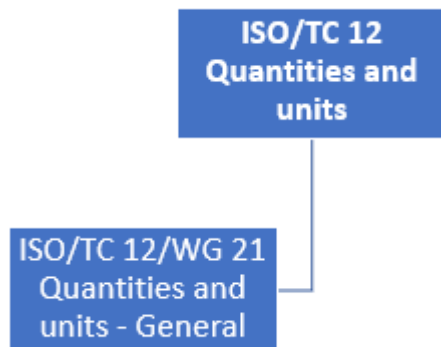


Meetings, held

- 1978 Copenhagen, Denmark
- 2010 Prague, Czech Republic
- 2012 Milan, Italy
- 2014 Berlin, Germany
- 2015 Gaithersburg, USA
- 2016 Stockholm, Sweden
- 2017 Berlin, Germany
- 2018-10-08, Finland
- 2020-05-28, Virtual via Zoom
- 2021-05-26, Virtual via Zoom

Meeting, planned 2022-05-19, Virtual via Zoom

Committee structure



ISO/TC 12 has one working group directly under the TC:

- **WG 21** “Quantities and units – General”, “*Revision of ISO 80000-1 “Quantities and units— Part 1: General”*”.
- Convenor: Prof. Leslie Pendrill*

Meeting held:

- 2020-02-27 via Zoom
- 2020-05-27, via Zoom
- 2021-01-21, via Zoom
- 2021-02-19, via Zoom
- 2021-03-19, via Zoom
- 2021-04-15, via Zoom
- 2021-05-11, via Zoom
- 2021-09-16, via Zoom

Business plan Current ISO/TC 12 Business plan was unanimously approved by ISO/TC 12, Draft resolution 140, 2020-04-09.
[STRATEGIC BUSINESS PLAN ISO/TC 12](#)

Programme of work and published documents

The standards produced by ISO/TC 12 are listed in Annex A.
[Work programme on ISO homepage](#)



Report and resolutions of the last ISO/TC 12 meeting

The resolutions taken and approved at the meeting are to be found in document [ISO/TC 12 N1425](#).

Follow-up on ISO/TC 12 Resolution 146-157 taken via Zoom meeting 2021-05-26

Resolution 149 (Zoom 5, 2021)

Agenda item 9.8 Revision of ISO 80000-1, *General*, Appointment of new Convenor

ISO/TC 12 decides to conduct a call for nominations for convenor of ISO/TC 12/WG 21 for the period from 2021-09-01 until 2024-08-31. ISO will then perform a Committee internal ballot to appoint the convenor.

P-members approving:	8
P-members disapproving:	0
P-members abstains:	0

The decision was unanimous

Resolution 150 (Zoom 6, 2021)

Agenda item 10 Study on the need for revisions in the ISO 80000 series Part 8, *Acoustics*

ISO/TC 12 decides to follow the recommendation TF1: 1:2021, Doc. N 1402:

Relationship to IEC 80000-15, Logarithmic quantities

It is decided that any revision needed of ISO 80000-1--12 proceeds without waiting for IEC WD 80000-15 *Logarithmic quantities and their units* to be finished. However, an exception, agreed with the PL of part 8, *Acoustics* was that IEC work on their part 15 should be completed before reviewing ISO 80000-8 since the two parts should be coordinated.

P-members approving:	8
P-members disapproving:	0
P-members abstains:	0

The decision was unanimous

**Resolution 151 (Zoom 7, 2021)**

Agenda item 10 Study on the need for revisions in the ISO 80000 series

ISO/TC 12 decides to follow the recommendation TF1: 4:2021, Doc. N 1402:

Corrected version of ISO 80000-2, Mathematics

ISO/TC 12/TF 1 decides to recommend ISO/TC 12 to prepare a corrected version that addresses the editorial comments and blatant errors in the document, see Table 1 in N1411. This excludes the comments from Finland (N1411 slide 17) which have a more technical character and will be left for the next SR (systematic review).

P-members approving:	8
P-members disapproving:	0
P-members abstains:	0

The decision was unanimous

Resolution 152 (Zoom 8, 2021)

Agenda item 10 Study on the need for revisions in the ISO 80000 series

ISO/TC 12 decides to confirm ISO 80000-3:2019, *Space and time*, and at the same time to ask ISO/CS to send the document for an earlier systematic review in October 2022.

It was noted that for part 3 *Space and time*, there were equal numbers of TF1 votes for doing a technical revision, or instead waiting for the next 5-year SR.

P-members approving:	7
P-members disapproving:	0
P-members abstains:	1 (DE)

The decision was passed by majority vote

**Resolution 153 (Zoom 9, 2021)**

Agenda item 10 Study on the need for revisions in the ISO 80000 series

ISO/TC 12 decides to follow the recommendation TF1: 5:2021, Doc. N 1402:
Minor revision of ISO 80000-9, Physical chemistry and nuclear physics, after systematic review

Register the comment in N1411, slide 34, and wait for the next SR.

P-members approving:	8
P-members disapproving:	0
P-members abstains:	0

The decision was unanimous

Resolution 154 (Zoom 10, 2021)

Agenda item 10 Study on the need for revisions in the ISO 80000 series

ISO/TC 12 decides to follow the recommendation TF1: 6:2021, Doc. N 1402:

Minor revision of ISO 80000-10, Atomic and nuclear physics, after systematic review

Register the comment in N1411, slide 35, and wait for the next SR.

P-members approving:	8
P-members disapproving:	0
P-members abstains:	0

The decision was unanimous

Resolution 155 (Zoom 11, 2021)

Agenda item 10 Study on the need for revisions in the ISO 80000 series

ISO/TC 12 decides to follow the recommendation TF1: 7:2021, Doc. N 1402:
ISO 80000-11, Characteristic numbers

Comments had been received (N1411 slide 36) and will wait for the next SR.

P-members approving:	8
P-members disapproving:	0
P-members abstains:	0

The decision was unanimous



Resolution 156 (Zoom 12, 2021)

Agenda item 10 Study on the need for revisions in the ISO 80000 series

ISO/TC 12 decides to follow the recommendation TF1: 8:2021, Doc. N 1402:

Corrected version of ISO 80000-12 condensed matter physics

There were editorial and blatant errors found (N1411 slides 37 - 39), which should be corrected in an updated version.

The comment from Prof. Obdržálek regarding 12-63.11 is of a technical nature and will be considered at the next SR.

P-members approving:	8
P-members disapproving:	0
P-members abstains:	0

The decision was unanimous

OK Resolution 157 (Zoom 13, 2021)

Agenda item 16 Requirements concerning a subsequent meeting

ISO/TC 12 decides to accept the kind offer from GOST R to host the next meeting of ISO/TC 12 in St. Petersburg, Russian Federation at the D.I. Mendeleyev Institute for Metrology, VNIIM, on 2022-05-19

P-members approving:	8
P-members disapproving:	0
P-members abstains:	0

The decision was unanimous

Follow up on Resolutions by correspondence 2021:

- Draft resolution 1-2021 CD (145) VIM 4
- Draft resolutions 2-13-2021 (146-157) meeting May 26, *see above*.
- Draft resolution 14/2021 (158) 4th CD OIML D5



ANNEX A

Standard and/or project under the direct responsibility of ISO/TC 12 Secretariat (5)

[International harmonized stage codes.](#)

[13 Published ISO standards](#)

STANDARD AND/OR PROJECT UNDER THE DIRECT RESPONSIBILITY OF ISO/TC 12 SECRETARIAT (14) ↓	STAGE	ICS
ISO 80000-1:2009 Quantities and units — Part 1: General	90.92	01.060
ISO 80000-1:2009/COR 1:2011 Quantities and units — Part 1: General — Technical Corrigendum 1	60.60	01.060
ISO 80000-2:2019 Quantities and units — Part 2: Mathematics	60.60	01.060
ISO 80000-3:2019 Quantities and units — Part 3: Space and time	60.60	01.060
ISO 80000-4:2019 Quantities and units — Part 4: Mechanics	60.60	01.060
ISO 80000-5:2019 Quantities and units — Part 5: Thermodynamics	60.60	01.060
IEC 80000-6:2008 Quantities and units — Part 6: Electromagnetism	90.92	01.060
ISO 80000-7:2019 Quantities and units — Part 7: Light and radiation	60.60	01.060
ISO 80000-8:2020 Quantities and units — Part 8: Acoustics	60.60	01.060
ISO 80000-9:2019 Quantities and units — Part 9: Physical chemistry and molecular physics	60.60	01.060
ISO 80000-10:2019 Quantities and units — Part 10: Atomic and nuclear physics	60.60	01.060
ISO 80000-11:2019 Quantities and units — Part 11: Atomic and nuclear physics	60.60	01.060
ISO 80000-12:2019 Quantities and units — Part 12: Condensed matter physics	60.60	01.060
IEC 80000-13:2008 Quantities and units — Part 13: Information science and technology	90.60	01.060

Joint work under IEC/TC 25 lead:

IEC 80000-6 *Quantities and units — Part 6: Electromagnetics* is under revision. Project leader: Prof. Göran Grimvall

IEC 80000-15 *Quantities and units — Part 15: Logarithmic and related quantities* is under development. Project leader: Prof. Jan Obdržálek.

IEC 80000-16 *Quantities and units — Part 16: Printing and writing rules* is under development. Project leader: Prof. Erik Jacobson.

IEC 80000-17 *Quantities and units — Part 17: Time dependency* is under development. Project leader: Prof. Erik Jacobson.



ANNEX B

P-members attending ISO/TC 12 meetings 2017-2021

Attending **Not attending**

P-member	2017 Berlin	2018 Finland	2020 Virtual	2021 Virtual
• Austria (ASI)				
• Belarus (BELST)				
• China (SAC)				
• Czech Republic (UNMZ)				
• Denmark (DS)				
• Egypt (EOS)				
• Finland (SFS)				
• France (AFNOR)				
• Germany (DIN)				
• India (BIS)				
• Italy (UNI)				
• Japan (JISC)				
• Kuwait (KOWSMD)				
• Portugal (IPQ)				
• Russian Federation (GOST R)				
• Spain (UNE)				
• Sweden (SIS)				
• Switzerland (SNV)				
• United Kingdom (BSI)				
• United States (ANSI)				